

Muhammad Abdul Rehman

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PROFESSIONAL SUMMARY

Computer Science graduate specializing in Data Science, with hands-on experience building AI-powered applications using Python, FastAPI, LangChain, TensorFlow, React, and modern LLM frameworks. Experienced in developing production-oriented RAG systems, local LLM deployments, multimodal AI solutions, REST APIs, and intelligent automation platforms. Passionate about AI Engineering, Machine Learning, NLP, Computer Vision, Generative AI and Software Development.

TECHNICAL SKILLS

- **AI / ML:** Generative AI, RAG, LLMs, NLP, Computer Vision, Machine Learning, Prompt Engineering, LLM Evaluation
- **Frameworks:** LangChain, Hugging Face, Groq, TensorFlow, Scikit-learn, OpenCV, LoRA, QLoRA
- **Programming & Web:** Python, JavaScript, FastAPI, Flask, REST APIs, React.js, Streamlit
- **Data / Databases:** Pandas, NumPy, BeautifulSoup, Scrapy, ETL Pipelines, Power BI, PostgreSQL, MongoDB, Pinecone, Firebase
- **Cloud / DevOps:** AWS (EC2), Azure AI Services, Docker, GitHub, Swagger, ReDoc

PROFESSIONAL EXPERIENCE

Data Scraping Intern | Programmers Force | Remote

July 2025 – Feb 2026

- Scraped and processed data from **2,000+ websites across 30+ countries** using a custom Chrome extension, configuring CSS/HTML tag selectors via browser DevTools (inspect mode) to extract structured data fields.
- Built entity-filtering pipelines to retain only records with verified person names; performed QA and data preprocessing, identifying malformed entries and escalating scraping failures with detailed error reports.

Full Stack Developer | Freelance | Remote

Oct 2025 – Present

- PTC Eightfold Autofill Chrome Extension: Built a Chrome extension (Manifest V3, chrome.storage.local, content/background scripts) that auto-fills job application form text fields, dropdowns, checkboxes, dates, and resume/cover-letter file uploads from saved or JSON-imported candidate data, eliminating repetitive manual entry.
- Event Management System: Built a REST API and CLI (Python, FastAPI, SQLAlchemy, SQLite) for a client to manage events and registrations, with full CRUD operations, concurrency-safe booking logic using SQLite WAL mode and locking to prevent overbooking, and a Rich-based interactive CLI as an alternative client to the API.
- Data Automation & Analytics: Designed advanced financial and expense tracking sheets for corporate clients utilizing complex Excel formulas, lookup functions, pivot tables, and data validation to automate manual budgeting workflows.

PROJECTS

IntraView AI (AI-Based Interview System) | Final Year Project

Sept 2025 – May 2026

Python, React, TypeScript, FastAPI, REST APIs, MongoDB, TensorFlow, Hugging Face, OpenCV, Docker

- Co-developed a **dual-role full-stack platform** (candidate + admin) utilizing FastAPI, React, TypeScript, and MongoDB with JWT authentication, role-based access control, and SMTP-based OTP password reset.
- Built a **2.5M-question interview bank** (coding, behavioural, system design) with difficulty controls and a **real-time Gemini API fallback** that generates questions on demand when no DB match is found.
- Built OpenCV + MediaPipe pipelines for real-time facial expression analysis; browser-based **Docker sandbox** for automated test-case grading; admin dashboard with live session monitoring and Swagger/ReDoc API docs.

Privacy Medical Chatbot

Feb 2026 – Apr 2026

Python, Meta Llama 3.1 8B, LangChain, Hugging Face, Pinecone, RAG, NLP, Flask, AWS EC2

- Deployed Meta Llama 3.1 8B locally via **Q4_K_M 4-bit quantization** (~4.58 GB), achieving **zero third-party API dependency** and full medical data privacy with a 2,048-token context window.
- Built RAG pipeline using LangChain, MiniLM-L6-v2 (384-dim embeddings), and Pinecone serverless search with 500-char PDF chunks; **top-2 retrieval in under 1 second** and custom hallucination reduction prompt template that strictly grounds LLM responses in retrieved context with source attribution, with 2 to 5 second end-to-end response times and supporting concurrent multi-user access.

AI Face Shape Detector & Stylist

Sep 2025 – Dec 2025

Python, TensorFlow, EfficientNetV2, CNN, OpenCV, MediaPipe, DeepFace, FastAPI

- Trained EfficientNetV2 via 2-stage transfer learning (frozen backbone then top-30% fine-tune with early stopping), improving validation accuracy from **60% to 83.4%**; hybrid classifier combining MediaPipe's **478-point landmarks** with confidence scores.
- Integrated DeepFace for **gender detection**, serving personalized recommendations (beard/hairstyle/glasses for males; makeup/hairstyle/glasses for females) via FastAPI, supporting live webcam and file upload.

EDUCATION

COMSATS University Islamabad | Pakistan

Sep 2022 – Jun 2026

BS Computer Science – Specialization: Data Science

CERTIFICATIONS

Generative AI with Large Language Models | DeepLearning.AI & AWS | Coursera

2025

ACTIVITIES

- Served as **Registration Club Co-Lead for ACM COMSATS**, coordinating student registrations for academic and tech events.
- Organized and managed events, including **VisioSpark Episode 20** and **Student Week**.